

5. (1998, B-5)

$$N = 1111 \dots 1$$

(1998 digits, all 1s)

Find the one thousandth decimal digit of $\sqrt{N}$.

We can write: $N = \sum_{k=0}^{1997} 10^k$ (Geometric Series)

Recall: $1 + r + r^2 + \dots + r^{n-1} + r^n = S_n$
 $r + r^2 + \dots + r^{n-1} + r^n + r^{n+1} = r S_n$

$$\Rightarrow 1 - r^{n+1} = S_n - r S_n = (1 - r) S_n \Rightarrow S_n = \frac{r^{n+1} - 1}{r - 1}$$

Thus $N = \sum_{k=0}^{1997} 10^k = \frac{10^{1998} - 1}{9} = N$

Now what? $\sqrt{N} = \frac{1}{3} \sqrt{10^{1998} - 1} = \frac{1}{3} \sqrt{(10^{999} + 1)(10^{999} - 1)}$

$$f(x) = \sqrt{c - x} = \sqrt{c \left(1 - \frac{x}{c}\right)} = \sqrt{c} \cdot \sqrt{1 - \frac{x}{c}}$$

$f(u) = \sqrt{c} \cdot \sqrt{1 - u}$ $\star$ $g(u) = \sqrt{1 - u}$ $\star$ call $u = \frac{x}{c}$

$$g(0) = 1$$

$$g'(0) = -\frac{1}{2} (1 - u)^{-1/2} = -\frac{1}{2}$$

$$g''(0) = -\frac{1 \cdot 1}{2 \cdot 2} (1 - u)^{-3/2} = -\frac{1}{4}$$

$$g'''(0) = -\frac{1 \cdot 1 \cdot 3}{2 \cdot 2 \cdot 2} (1 - u)^{-5/2} = -\frac{3}{8}$$

$$g(u) \approx 1 - \frac{1}{2}u - \frac{1}{4 \cdot 2!}u^2 - \frac{3}{8 \cdot 3!}u^3 - \dots$$

$$f(x) \approx \sqrt{c} \left[1 - \frac{x}{2c} - \frac{x^2}{8c^2} - \dots \right]$$

$$g^{(n)}(0) = -\frac{(2n-3)!!}{2^n} (1 - u)^{-(2n-1)/2} = -\frac{(2n-3)!!}{2^n}$$

$$f(x) \approx \sqrt{c} \left[1 - \frac{x}{2c} - \frac{x^2}{4c^2} - \dots \right]$$

$$= 10^{999} \left[1 - \frac{1}{2} \cdot 10^{-1992} - \frac{1}{4} \cdot 10^{-2996} - \dots \right]$$

Likely insignificant... need to prove.

$$\approx 10^{999} - \frac{1}{2} 10^{-999}$$

↳ Due to geometric progression will never affect above say, 2000th decimal places

Therefore,

$$\sqrt{N} \approx \frac{1}{3} \left(10^{999} - \frac{1}{2} 10^{-999} \right) \quad \left\{ \begin{array}{l} \text{with acceptable} \\ \text{error.} \end{array} \right.$$

$$= \underbrace{\frac{1}{3} \cdot 10^{999}}_{\text{Term \#1}} - \underbrace{\frac{1}{6} 10^{-999}}_{\text{Term \#2}}$$

Term 1 $\boxed{333 \dots 33.33\overline{3}}$ (repeating 3s)

Term 2 10^{-999} will be $0.\underbrace{000 \dots 01}_{998 \text{ zeros}}$

So, $\frac{1}{6} 10^{-999} = 0.\underbrace{000 \dots 0016\overline{6}}_{999 \text{ zeros}}$

$10^{-1} = 0.1$
 $10^{-2} = 0.00\overline{1}$

So we have

...	3	3	3	3	3	...	3	3	3	3	3
	0	0	0	0	...	0	0	1	6	6	7

← not 7!

... 3 3 3.3 3 3 ... 3 3 1 6 6 6

So, we will have a 1 in the thousandth decimal place. □